

SMART PARKING PLATFORM

Homework 2: Work Breakdown Structure
and Project Timeline

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4. Work Breakdown Structure (WBS)

The WBS decomposes the Smart Parking Platform into progressively smaller components. It is organized by product features at level 2, authentication, user & operator management, dashboards, reporting & analytics, mapping & notifications, platform & integrations, and project management, with individual work packages at level 3 and a further breakdown at level 4.

4.1 WBS Overview Diagram

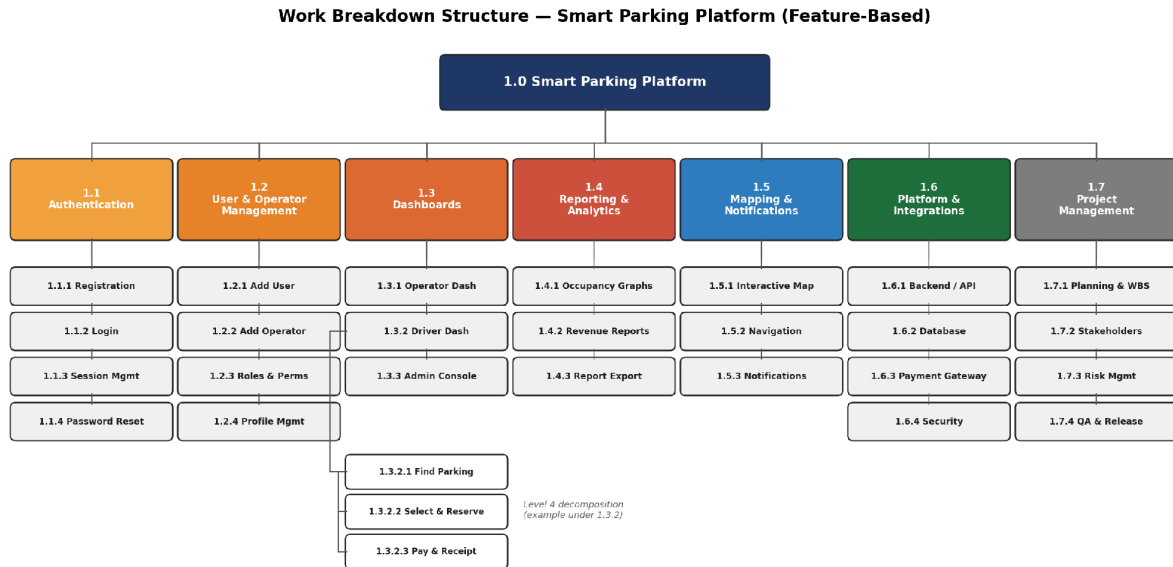


Figure 1: WBS for the Smart Parking Platform, level 2 groups the platform by feature area, level 3 lists work packages; the driver dashboard is shown decomposed to Level 4 as an example. The full decomposition appears in Table 1.

4.2 WBS Dictionary

The table below lists each WBS element with its deliverable and owning team, level 1 is the overall project, level 2 represents feature areas, level 3 represents the work packages that will be scheduled, and level 4 (shown under the Dashboards feature) shows the finest level of decomposition used on this project.

WBS ID	Work Package / Element	Deliverable	Owner
1.0	Smart Parking Platform	Complete platform delivered to operators and drivers	All Teams
1.1	Authentication	Secure account access across web and mobile	Backend / Mobile / Web
1.1.1	Registration	New account sign-up with email/phone verification	Backend/API
1.1.2	Login	Credential-based sign-in for all user roles	Backend/API
1.1.3	Session Management	Token issue/refresh, timeout, secure sign-out	Backend/API
1.1.4	Password Reset & Recovery	Self-service password reset flow	Backend/API
1.2	User & Operator Management	Account administration for all user types	Backend / Web
1.2.1	Add User	Create and manage driver accounts	Backend/API
1.2.2	Add Operator	Onboard parking operators and their facilities	Web App
1.2.3	Roles & Permissions	Role-based access control across the platform	Backend/API
1.2.4	Profile Management	User-editable profile, vehicle, and payment details	Mobile / Web
1.3	Dashboards	Primary interfaces for drivers and operators	Mobile / Web
1.3.1	Operator Dashboard	Garage management and live occupancy view	Web App
1.3.1.1	Add / Edit / Remove Garage	Facility CRUD: address, capacity, hours, rules	Web App
1.3.1.2	Pricing Configuration	Base rates and time-based pricing rules	Web App
1.3.1.3	Live Occupancy View	Real-time facility occupancy display	Web App
1.3.2	Driver Dashboard	Find, reserve, and pay for parking	Mobile / Web
1.3.2.1	Find Parking	Search and filter available spaces near a destination	Mobile / Web
1.3.2.2	Select & Reserve	Space selection and reservation hold	Mobile / Web
1.3.2.3	Pay & Receipt	Payment capture, confirmation, and receipt	Payment
1.3.3	Admin Console	Platform-wide user, role, and integration management	Web App
1.4	Reporting & Analytics	Operator-facing reporting and data export	Web App
1.4.1	Occupancy Graphs	Utilization charts over selectable date ranges	Web App
1.4.2	Revenue Reports	Financial summaries by facility and period	Web App

WBS ID	Work Package / Element	Deliverable	Owner
1.4.3	Report Export	Export financial and occupancy reports (CSV/PDF)	Web App
1.5	Mapping & Notifications	Location services and user messaging	Mapping / Mobile
1.5.1	Interactive Map	Live availability layer over a map view	Mapping
1.5.2	Navigation Hand-off	Turn-by-turn routing to the reserved facility	Mapping
1.5.3	Notifications & Alerts	Confirmation, expiration, and change alerts	Mobile App
1.6	Platform & Integrations	Backend foundation and third-party services	Backend / Payment
1.6.1	Backend / API Development	Core services and documented REST API contract	Backend/API
1.6.2	Database	Data model, schema, and persistence layer	Backend/API
1.6.3	Payment Gateway Integration	Third-party payment processing and refunds	Payment
1.6.4	Security & Compliance	Data privacy, security controls, city ordinance rules	Backend/API
1.7	Project Management	Plans, schedules, reports, and release control	PM Office
1.7.1	Planning & WBS	Project charter, WBS, schedule baseline	PM Office
1.7.2	Stakeholder Management	Stakeholder register, weekly updates, communications	PM Office
1.7.3	Risk Management	Risk register and mitigation plans	PM Office
1.7.4	QA, Testing & Release	Test plan, defect log, UAT sign-off, deployment	QA & Testing

Table 1: WBS dictionary — elements, deliverables, and owning teams across four levels of decomposition.

5. Project Timeline

The schedule below maps the WBS feature areas onto the fixed semester timeline.

5.1 Gantt Chart

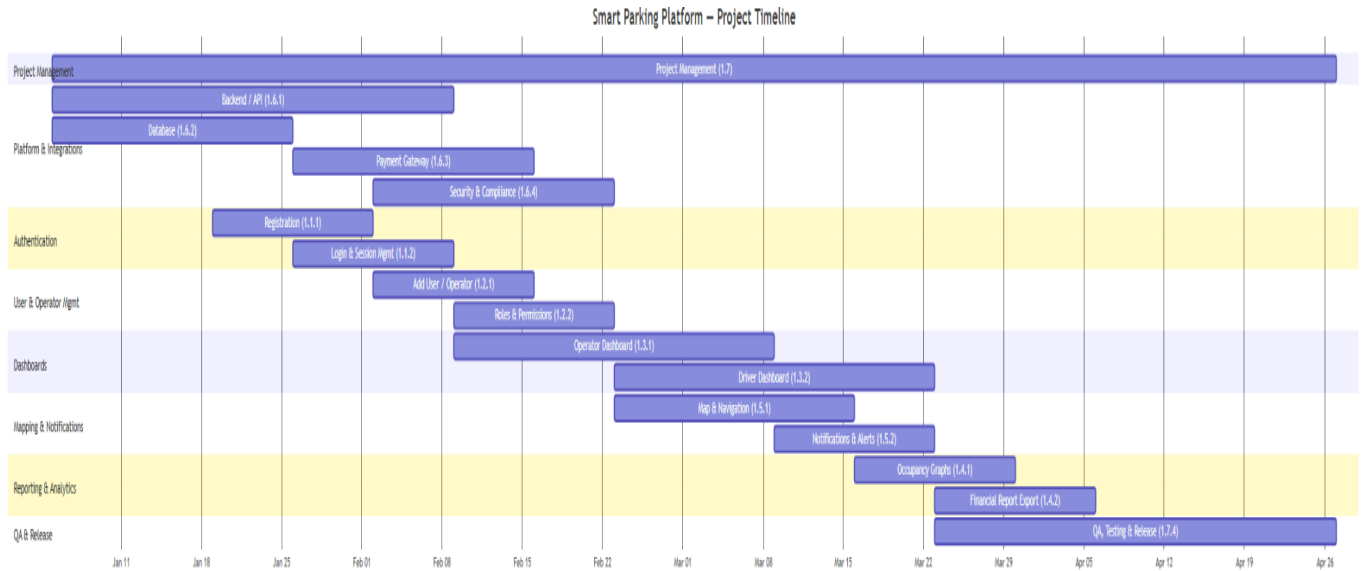


Figure 2: Gantt chart showing WBS feature areas and work packages across the 16-week semester.